APELEVEL / ENGINEERING REPORT

Whisper Market Treasury Funding & Live Balance Verification

Polygon Amoy testnet | October 28, 2026 | RAGE REALMZ x Whisper Oracle

Executive summary

ApeLevel deployed the Whisper Market Treasury smart contract on Polygon Amoy and funded it with 100,000 existing tTILT. The funding was completed through its dedicated wallet-signing portal. Subsequent independent RPC calls confirmed a live ERC-20 treasury balance of 100,000.0 tTILT at block 51,375,947. The RAGE REALMZ marketplace separately displayed the player's own tTILT balance and the live treasury balance, as shown by the attached interface evidence.

MEASURE	VERIFIED / REPORTED RESULT
Treasury balance	100,000.0 tTILT (100000000000000000000000 base units)
Amoy network	Chain ID 80002
Treasury contract	0x58a4Cc95F89E52bB7965FDf55F5C91937ED17043
Test token	0xc1dd560375459b19d6437f076f1708e286bb5d67
Resources contract	0x2Ac9bC76958Add0cA49D534de3e6d9D7c2ba3eBD
Treasury tests	21 passing; 0 failures after spending-limit change

What this milestone proves

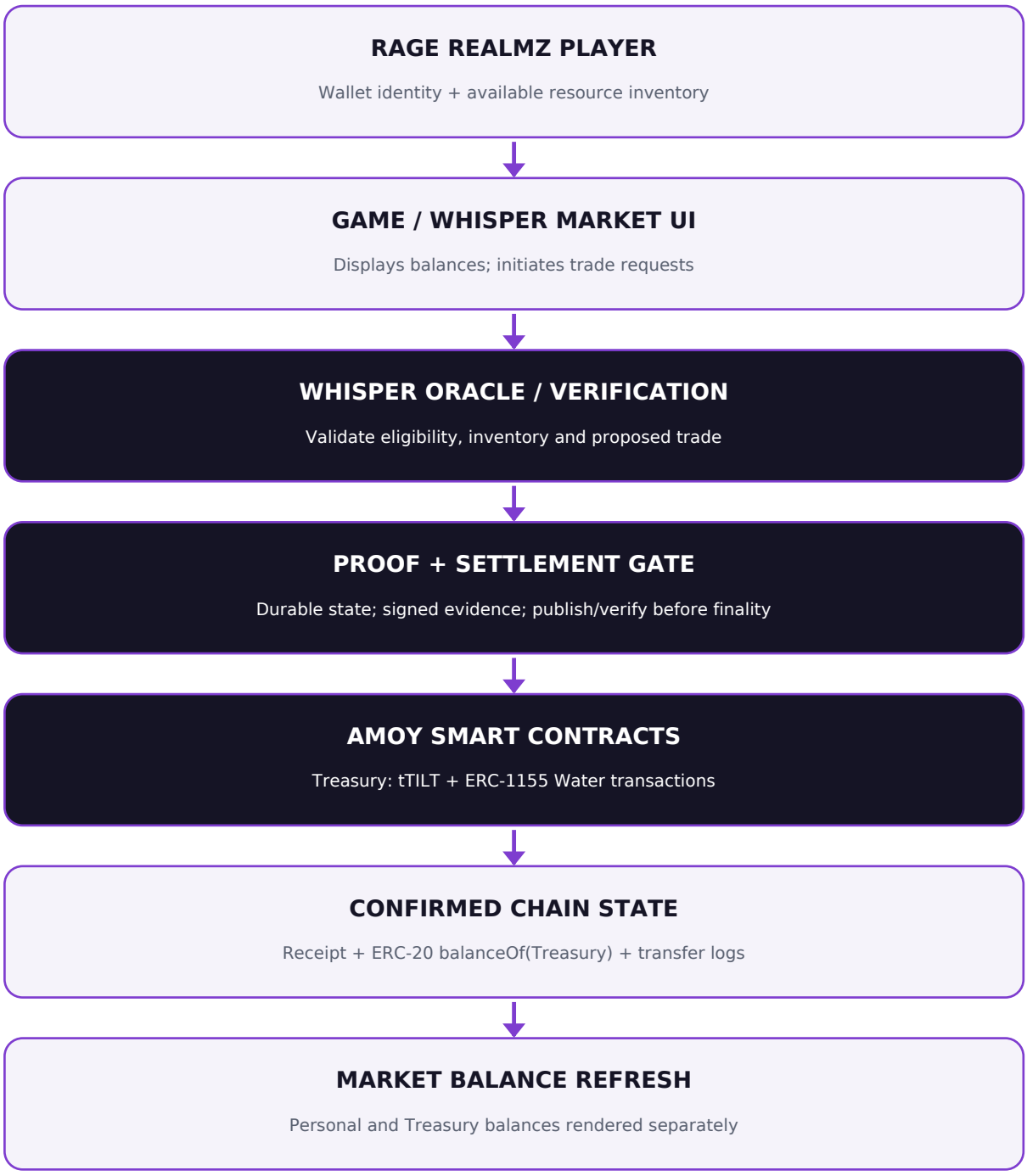
The screenshots and RPC test together support three findings: (1) the funding workflow reported a confirmed Treasury deposit; (2) an independent read of the deployed ERC-20 returned the 100,000 tTILT balance, without using a hardcoded display value; and (3) the in-game Whisper Market rendered that treasury value separately from the connected player's balance. These results do not yet establish that the full Oracle resource-trading and settlement pipeline is operational.

Why ApeLevel engineered Whisper Oracle

RAGE REALMZ manages player activity and resources, while tTILT payments occur on a public blockchain. ApeLevel designed Whisper to bridge these environments without treating a client claim, marketplace click, or wallet popup as final proof of payment or delivery.

The intended design combines durable transaction records, checks of wallet and resource ownership, atomic reservation/reconciliation, controlled authorization, signed and sanitized trade evidence, a public proof-publication gate, and blockchain receipt verification. The actual treasury smart contract currently trades ERC-1155 Water; reconciliation with the game's off-chain inventory remains to be implemented and verified.

Proposed end-to-end operating flow



Flowchart depicts the intended complete architecture. The on-chain funding and read-only balance/display stages are demonstrated; full Oracle proof gating and inventory reconciliation are not established by the screenshots.

Implementation and test evidence

1. Treasury and Resources contracts

The deployed WhisperMarketTreasury contract holds an immutable tTILT token reference and an immutable ERC-1155 Resources reference. Its deposit(uint256) function is restricted to designated ApeLevel wallets, transfers approved tTILT using SafeERC20, records depositor totals and emits a TreasuryDeposit event. Its Water trading methods require sufficient balances, resource approval, and include ReentrancyGuard protection.

The original 1,000 tTILT lifetime Water-purchase budget was intentionally removed. The 100-Water per-transaction restriction on treasury purchases remains. The resulting Foundry treasury suite passed 21 of 21 tests, including deposit authorization, failed-purchase behavior, reentrancy rejection, inventory restrictions and Water buying/selling in the isolated contract tests.

2. Portal-controlled funding

ApeLevel reused its isolated Codespaces deployment portal on port 4173. The interface was configured to verify Polygon Amoy, the authorized wallet, the token, and the Treasury address. It requested ERC-20 approval followed by the Treasury deposit transaction. The portal then displayed a confirmed deposit and a 100,000 tTILT treasury balance. Wallet approval and the on-chain contract, rather than the webpage alone, govern actual token movement.

3. Independent blockchain verification

A separate ethers.js verification script connected to Polygon Amoy RPC and called balanceOf(Treasury) directly on the deployed tTILT token. It also queried the Treasury's immutable token and Resources references. The returned balance was 100000.0 tTILT at block 51,375,947, matching the expected funding amount. This validation did not consume a local balance record or a UI constant.

4. In-game marketplace display

The game on port 3000 has distinct MY tTILT BALANCE and WHISPER TREASURY BALANCE fields. The treasury display uses an Amoy eth_call of ERC-20 balanceOf against the fixed deployed Treasury address, refreshes when the market opens, and polls approximately every eight seconds while open. An event hook was prepared for confirmed Whisper transactions, but a production Oracle emitter is not yet shown.

Public proof references

Treasury contract on Amoy Polygonscan

tTILT token and balance / transfer records

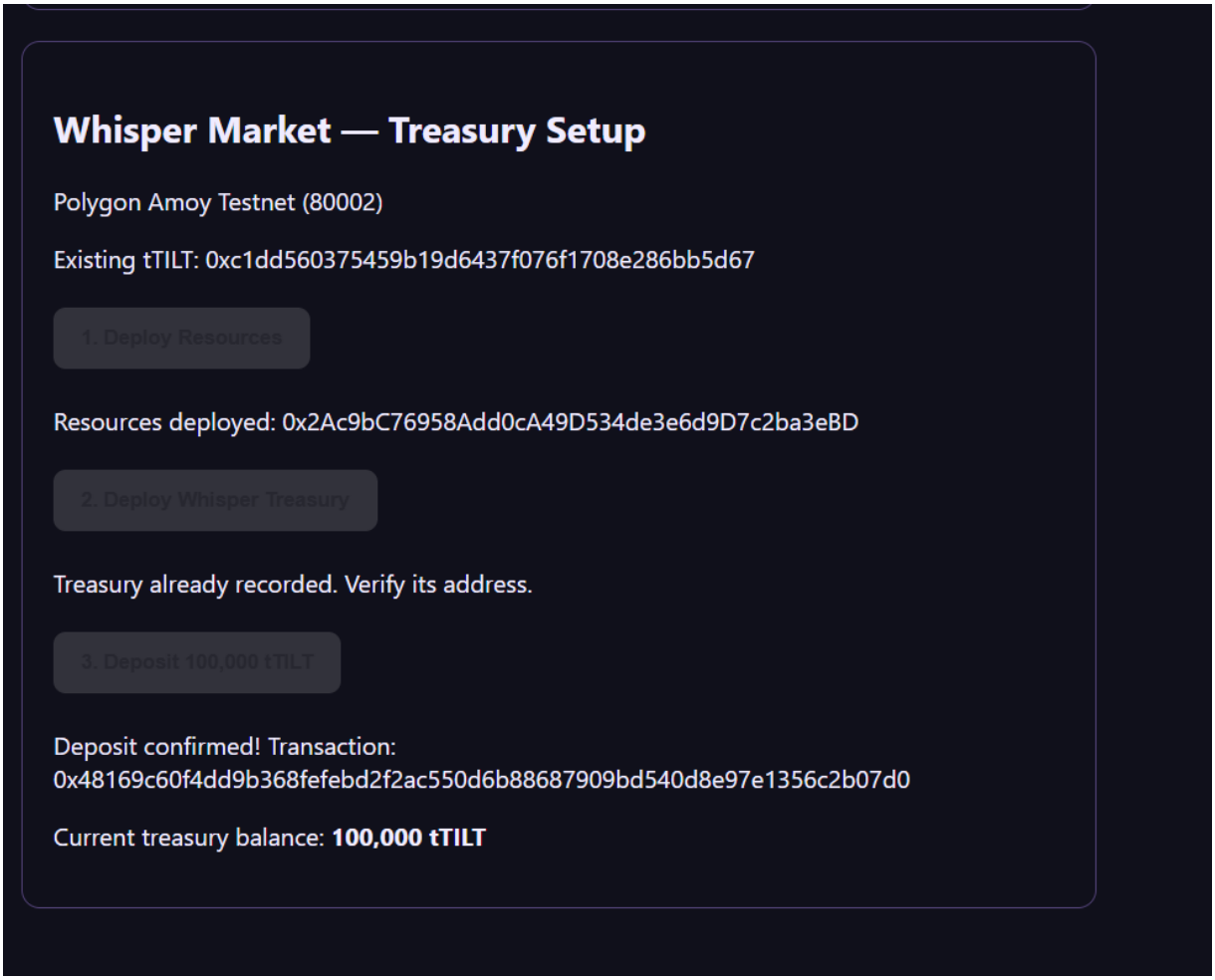
Treasury deployment transaction

ApeLevel Whisper public repository

The deposit transaction hash displayed by the provided portal screenshot is reproduced in the figure on the next page. To certify individual transaction events independently, review the explorer receipt and token Transfer logs.

Figure 1 - ApeLevel deployment portal: Treasury funded

The custom deployment portal displays both deployed contracts, a deposit-confirmed message, the funding transaction identifier and a current treasury balance of 100,000 tTILT.



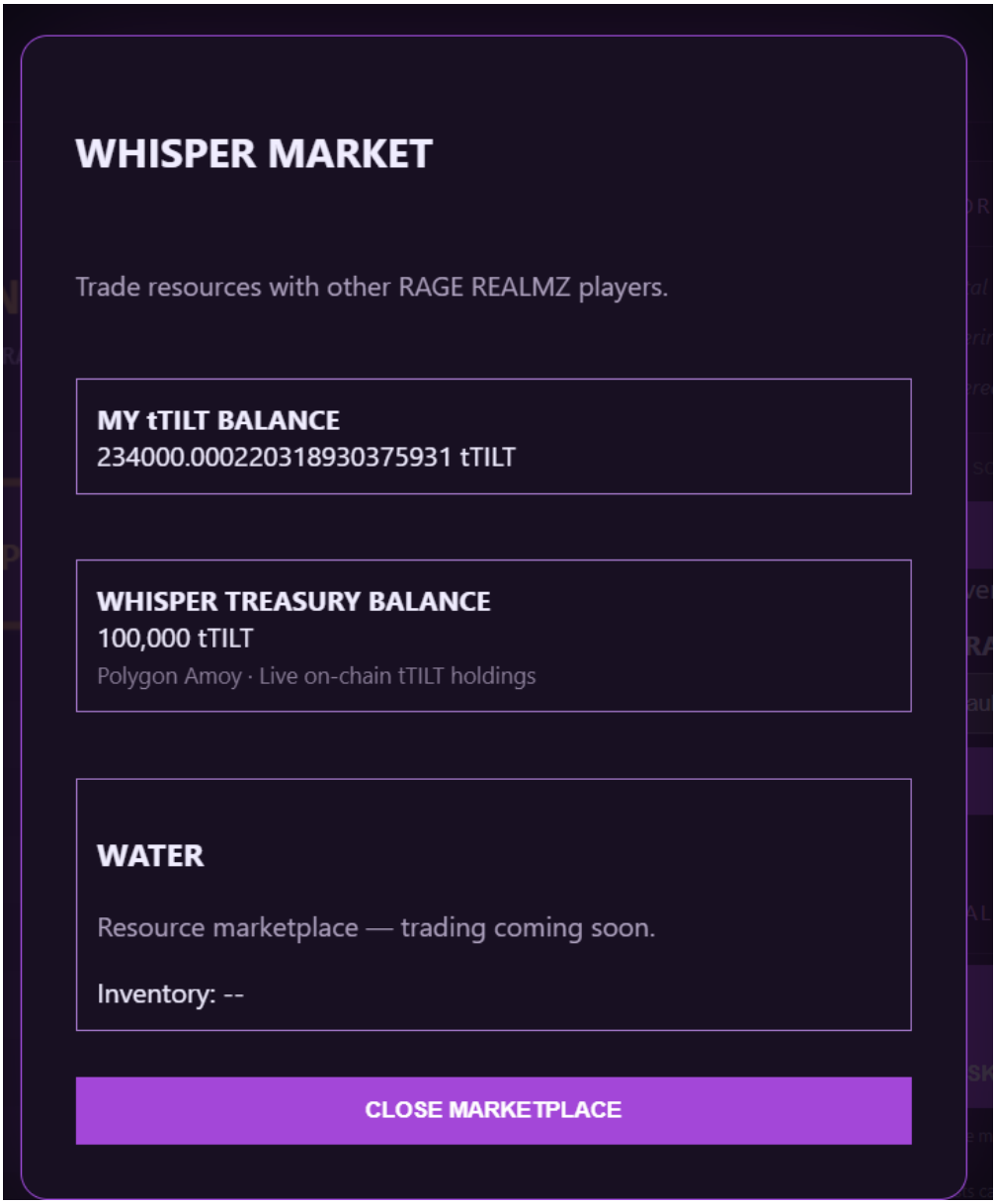
Source: user-supplied screenshot of the ApeLevel Amoy deployment portal. The image establishes what the portal showed at capture time; the separate RPC verification establishes the on-chain balance.

Evidence interpretation

An interface confirmation alone would not be enough to conclude that funds are held by the intended contract. Here, the separate live balance call returned the same 100,000 tTILT quantity and verified the linked token and resource addresses. The two types of evidence therefore reinforce one another.

Figure 2 - Whisper Market inside RAGE REALMZ

The game marketplace shows two distinct on-chain balance areas. At capture, the player balance displayed 234000.000220318930375931 tTILT, while the Treasury displayed 100,000 tTILT. The Water section remained marked as trading coming soon.



Source: user-supplied in-game screenshot. Live balance values may change with confirmed chain activity. The screenshot does not demonstrate a successful resource trade.

What is proven - and what remains

STATUS	ENGINEERING CLAIM
Demonstrated	Treasury contract deployed and linked to the intended tTILT and ERC-1155 contracts
Demonstrated	100,000 tTILT confirmed by independent balanceOf query at block 51,375,947
Demonstrated	Game UI distinguishes personal balance from live treasury holdings
Tested locally	21 Treasury Foundry tests passed, including buy/sell and reentrancy test cases
Not yet demonstrated	Live player resource trade settled end-to-end through Whisper Oracle
Not yet demonstrated	Production Oracle-signed proof published and reconciled before every trade finalization
Not yet demonstrated	Synchronization between actual gameplay inventory and ERC-1155 Water holdings

Why the architecture matters

ApeLevel intends to finance resource liquidity with a treasury whose token holdings are independently checkable. Players will eventually be able to exchange valid in-game resources for tTILT, subject to resource ownership, approved transfers, contract balances and Oracle reconciliation. A separately visible Treasury balance improves transparency: users can see the capacity available to fund resource purchases rather than relying on a claimed off-chain reserve.

The treasury has no cumulative lifetime spending ceiling. Instead, its buyWater method requires sufficient available tTILT and uses transaction-level protections. Treasury inventory similarly constrains the amount of Water it can sell. This keeps basic market operations tied to actual asset holdings, while the planned Oracle layer must enforce game-economy validity and cross-system consistency.

Next engineering acceptance tests

Before enabling player trades: confirm the TreasuryDeposit and Transfer events for funding; implement and test the game-inventory to ERC-1155 reconciliation path; exercise a small on-chain Water sale/purchase with a controlled test wallet; require receipt finality and public Oracle proof verification; test retries, disconnects, chain changes and failed transactions; and verify that the UI converges to the same confirmed token balances after each settlement.

Conclusion

The initial funding and live marketplace display are real, independently corroborated Amoy testnet milestones. They demonstrate that ApeLevel can deploy a restricted treasury, fund it with existing testnet tokens, read its live on-chain holdings and present that balance inside RAGE REALMZ. The next milestone is to prove a complete, auditable resource trade through Whisper Oracle, rather than implying that the market is already production-operational.